



AYE  
TECH HUB

• MEP ENGINEERING • LESSON 03

# Reading MEP Drawings and Specifications

Symbols · Numbering · Scales · Title Blocks · Plans

Lesson 03 of 30 | MEP Engineering Program

**Awet G. Nway**

Founder, AYE Tech Hub | MEP & Building Services Engineer

- The 5 levels of MEP drawing from concept to as-built
- MEP drawing numbering and cross-referencing systems
- Common mechanical, electrical and plumbing symbols
- How to read a mechanical floor plan and ductwork layout
- Title blocks, revision clouds, and drawing registers

# Why MEP Drawings Are the Language of Construction

Every piece of ductwork, every cable tray, every pipe run in a building was installed exactly as shown on a drawing. MEP drawings are the primary communication tool between designers, contractors, subcontractors, inspectors, and building owners. If you cannot read them accurately, you cannot design, install, inspect, or maintain building services.

Reading MEP drawings is a skill — one that takes practice to develop. The goal of this lesson is to give you the **framework for reading any MEP drawing set systematically**, regardless of which project, which discipline, or which country's conventions the drawing follows.

## The 5 Levels of MEP Drawing

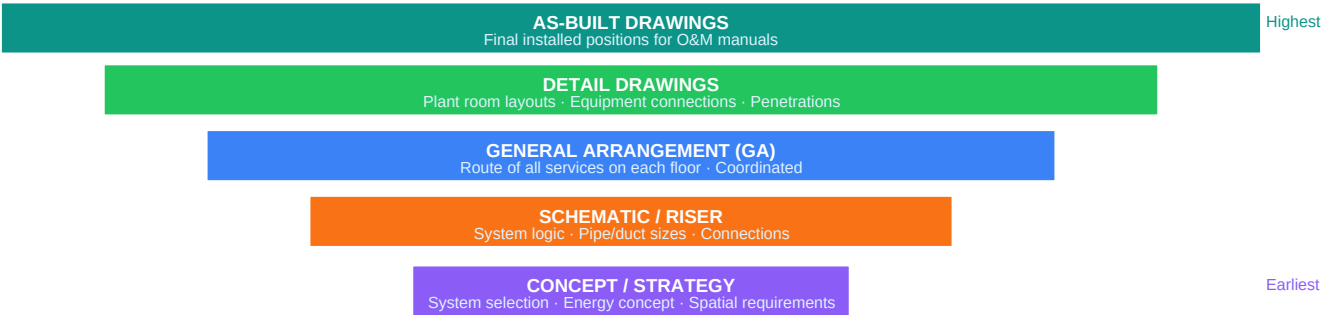


Figure 1 — MEP Drawing Hierarchy: From Concept to As-Built

| Drawing Type              | Scale (typical) | What to Look For                                                         | Audience                                                      |
|---------------------------|-----------------|--------------------------------------------------------------------------|---------------------------------------------------------------|
| Concept / Strategy        | Not to scale    | System types, energy source, plant room locations, riser strategy        | Client, architect, cost consultant                            |
| Schematic / Riser Diagram | Not to scale    | System logic, component connections, pipe/duct sizes, flow rates         | Designer, contractor, commissioning engineer                  |
| General Arrangement (GA)  | 1:100 or 1:50   | Routing of all services, spatial coordination, access provisions         | All contractors — primary installation drawing                |
| Detail Drawing            | 1:20 or 1:10    | Plant room layout, equipment connections, support details, fire stopping | Specialist subcontractors, installation teams                 |
| As-Built Drawing          | Same as GA      | Actual installed positions vs designed positions — mark-up of changes    | Facilities manager, O&M; team, future maintenance contractors |

*Always start with the drawing register: Every professional drawing set includes a drawing register (or index) — a list of all drawings in the package with their number, revision, title and status. Start here to understand the full scope of the package before reading individual sheets. Never assume a drawing is the latest revision without checking the register.*

# Drawing Numbering, Referencing and Symbols

Every drawing in a professional set has a unique number. Understanding the numbering convention lets you instantly identify the discipline, building level, and drawing type from the number alone — without even looking at the sheet.

| Numbering Component | Example                                                            | What It Means                                                                        |
|---------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Discipline prefix   | M = Mechanical E = Electrical P = Plumbing/Drainage                | First letter(s) identifies which MEP discipline the drawing covers                   |
| Level / zone code   | B1 = Basement 1 GF = Ground Floor L03 = Level 3 RF = Roof          | Specifies which floor or zone of the building the drawing represents                 |
| Drawing type code   | 100 = Floor plan / GA 200 = Schematic 300 = Detail 400 = Section   | Number range indicates the type of information on the drawing                        |
| Sequential number   | -01, -02, -03...                                                   | Multiple drawings of the same type and level are numbered in sequence                |
| Revision suffix     | Rev A = first issue Rev B = first revision Rev C = second revision | Every change to a drawing gets a new revision letter. Always use the latest.         |
| Full example        | M-L03-100-02 Rev C                                                 | Mechanical (M), Level 3 (L03), GA Plan (100), Sheet 2 of series, Third issue (Rev C) |

## Common MEP Drawing Symbols

| MECHANICAL                 | ELECTRICAL                | PLUMBING                  |
|----------------------------|---------------------------|---------------------------|
| ● Supply air duct (rect.)  | ● Distribution board (DB) | ● Cold water supply (CWS) |
| ● Return air duct          | ● Single socket outlet    | ● Hot water supply (HWS)  |
| ● Fan coil unit (FCU)      | ● Fluorescent luminaire   | ● Soil/waste pipe         |
| ● AHU (Air Handling Unit)  | ● Emergency lighting      | ● Vent/overflow pipe      |
| ● Chilled water flow (CHW) | ● Fire alarm call point   | ● Floor drain (FD)        |
| ● Low Temp HW flow (LTHW)  | ● RCD/RCBO consumer unit  | ● Sprinkler head          |

Figure 2 — Common MEP Drawing Symbols by Discipline

| Symbol Convention            | Mechanical                              | Electrical                              | Plumbing                                 |
|------------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------|
| Line weight (heavy)          | Main duct / primary pipe                | Power distribution cable                | Main drain / soil stack                  |
| Line weight (medium)         | Branch ductwork / secondary pipe        | Sub-circuit wiring                      | Branch waste pipe                        |
| Line type (dashed)           | Concealed duct above ceiling            | Low voltage / data cabling              | Concealed pipe / future works            |
| Line type (dot-dash)         | Centreline of duct or pipe              | Earth conductor                         | Centre of drain                          |
| Colour (if coloured drawing) | Blue=CHW, Red=HHW, Green=Gas, Grey=Duct | Red=power, Blue=data/comms, Green=earth | Blue=CWS, Red=HWS, Brown=soil, Grey=vent |

# The Title Block — Your First Stop on Any Drawing

The title block is found in the bottom-right corner of every professional drawing. It contains all the administrative information you need to verify you have the right drawing: the correct project, the correct revision, and the correct discipline.

| DRAWING TITLE BLOCK                                     |                                      |
|---------------------------------------------------------|--------------------------------------|
| Project Name:<br>Addis Heights Commercial Tower         | Client:<br>Addis Development PLC     |
| Drawing Title:<br>Level 3 — Mechanical Services GA Plan | Consultant:<br>AYE MEP Consultants   |
| Drawing Number:<br>M-03-100 Rev C                       | Checked By:<br>A.G. Nway (HMEP)      |
| Scale:<br>1:50 (A1)                                     | Approved By:<br>T. Bekele (Director) |
| Date:<br>2026-05-15                                     | Sheet:<br>3 of 28                    |

Figure 3 — MEP Drawing Title Block: Key Fields and Typical Information

| Title Block Field         | What to Check                                                                    | Red Flag                                                           |
|---------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Drawing Number            | Does it match the drawing register? Is this the number you requested?            | Number does not match register — you may have the wrong drawing    |
| Revision                  | Is this the latest revision per the drawing register?                            | Earlier revision than what is in the register — superseded drawing |
| Scale                     | Match scale bar to stated scale. Use on A1 originals only — photocopies distort. | No scale bar — never scale from this drawing                       |
| Date                      | Compare to project programme — is this drawing recent enough to be valid?        | Drawing dated before major design change — may be out of date      |
| Checked and Approved      | Must be signed off. Un-checked drawings should not be issued for construction.   | No checker/approver signature — not an issued document             |
| Status / Purpose of Issue | For Construction (FC) vs For Comment (FCA) vs For Information (FI)               | FI or FCA issued to site — work should not start from these        |

## How to Read a MEP Floor Plan — Step by Step

| Step | Action                                                          | What You Learn                                                                    |
|------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1    | Check title block — confirm discipline, level, revision, status | Verify you have the right drawing before spending time on it                      |
| 2    | Read the legend / key — find all symbols used on this drawing   | Understand what every line type, hatching and symbol means                        |
| 3    | Identify the north arrow and grid lines                         | Orientate yourself to the building plan and understand column grid references     |
| 4    | Locate all plant rooms, risers and service routes               | Understand where services enter the floor and how they distribute                 |
| 5    | Follow primary routes first, then branches                      | Understand the system hierarchy: primary mains first, then secondary distribution |
| 6    | Check for coordination notes and clash symbols                  | Identify unresolved clashes or coordination issues that need attention            |
| 7    | Read all text notes and schedules on the drawing                | Equipment tags, pipe/duct sizes, invert levels, and special requirements          |
| 8    | Cross-reference to schematics and details                       | Use drawing cross-references to understand system logic and component specs       |

*What comes next — MEP-004: HVAC Fundamentals for MEP — the mechanical side of building services in depth: HVAC system types, load calculation principles, ductwork sizing, AHU selection, and how HVAC integrates with the electrical and plumbing systems in a coordinated building services design.*